

# HexCode – AI-powered Portfolio Generator [DA 214 Course Project]

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## 1. Project Overview

HexCode is an AI-driven platform designed to simplify the creation of personalized portfolio websites. It empowers users—whether students, job seekers, or professionals—to transform their personal and professional details into a polished, responsive, and deployable portfolio without writing a single line of code.

*Transform your ideas into a beautiful, deployable portfolio using natural language.*

## 2. Objectives

- Enable effortless portfolio generation through natural language prompts.
- Provide customizable design options including layout and color palette.
- Support secure login and real-time collaborative editing.
- Facilitate easy one-click deployment and code export options.

## 3. Features and Capabilities

### Core Features

- AI-Driven Layout Generator
- Google OAuth Authentication
- Real-time Collaboration

### User Experience

- Form-driven input for About, Projects, Skills, etc.
- Drag-and-drop layout customization
- Live preview of edits

### Theme Personalization

- Multiple color palettes for user-specific branding

## Deployment & Code Access

- Instant Deployment
- Code Export

## 4. Tech Stack

| Layer          | Technologies Used                                  |
|----------------|----------------------------------------------------|
| Frontend       | React.js, framer-motion, Tailwind CSS, SweetAlert2 |
| Backend        | Node.js, Express.js, MongoDB                       |
| AI Integration | Anthropic Claude API                               |
| Authentication | Google OAuth 2.0                                   |

## 5. Setup & Installation Guide

### Prerequisites

- Node.js (v14+), npm (v6+)
- MongoDB (v4.4+)
- Google Cloud Platform
- Anthropic API Access

### Installation Steps

#### 1. Clone Repository

```
git clone https://github.com/Vibha17/CV_creator
```

#### 2. Navigate to Project Folder

```
cd CV_creator
```

#### 3. Install Frontend & Backend Dependencies

```
cd frontend && npm install
```

```
cd ../backend && npm install
```

4. Configure Environment Files (.env)
5. Start the Development Servers

## **6. Functional Workflow**

### **Step 1: Login with Google**

- Users authenticate using Google OAuth 2.0, securing their session.

### **Step 2: Fill Prompt Areas**

- Users provide details like:
  - Name, Profile Picture
  - About
  - Projects & Skills
  - Extracurriculars

### **Step 3: AI Prompt Processing**

- Claude API processes the input and generates a portfolio layout.

### **Step 4: Portfolio Generation**

- Responsive website is generated based on user details.

### **Step 5: Customization**

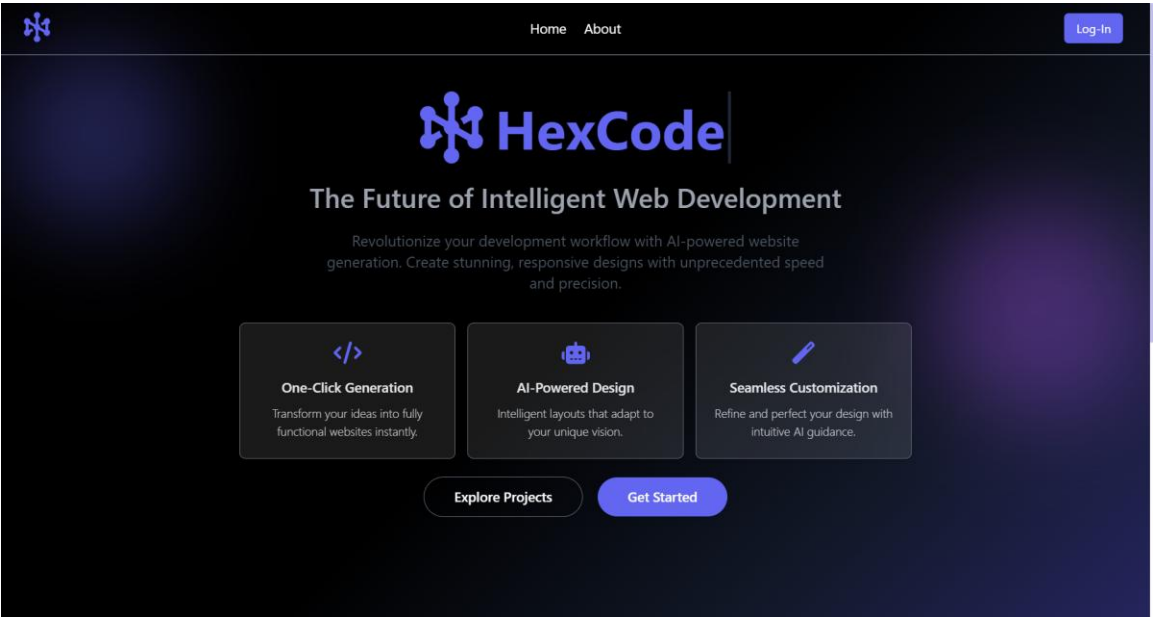
- Users can:
  - Edit layouts using drag-and-drop
  - Choose color themes
  - Preview real-time updates

### **Step 6: Collaborate & Deploy**

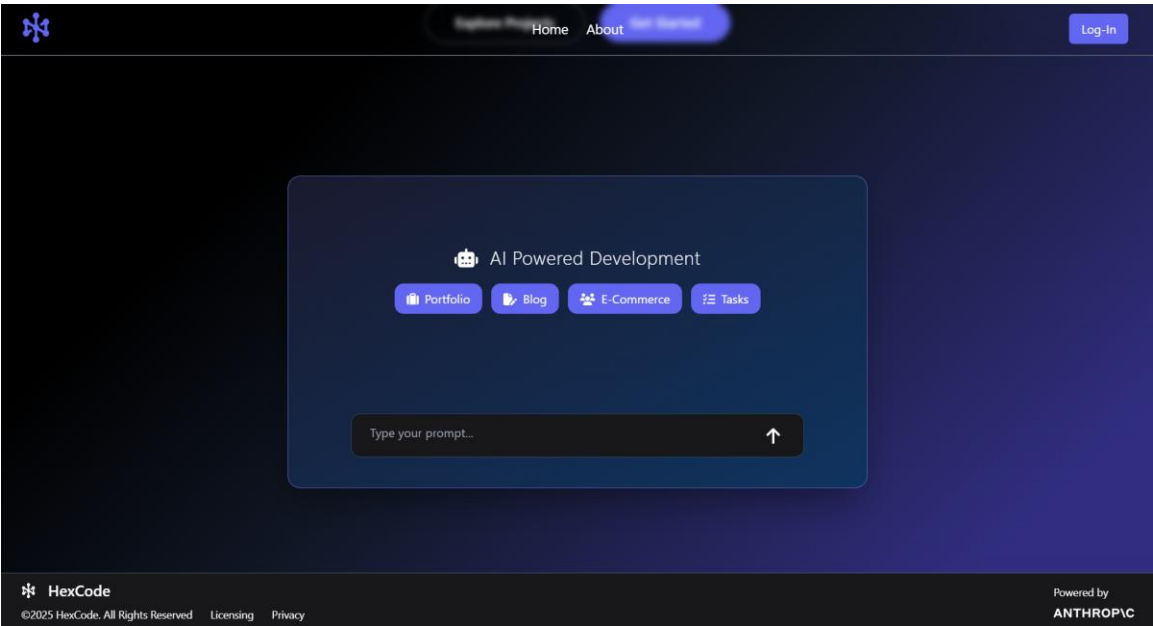
- Invite team members, share links, deploy live, or export code.

# 7. Screenshots

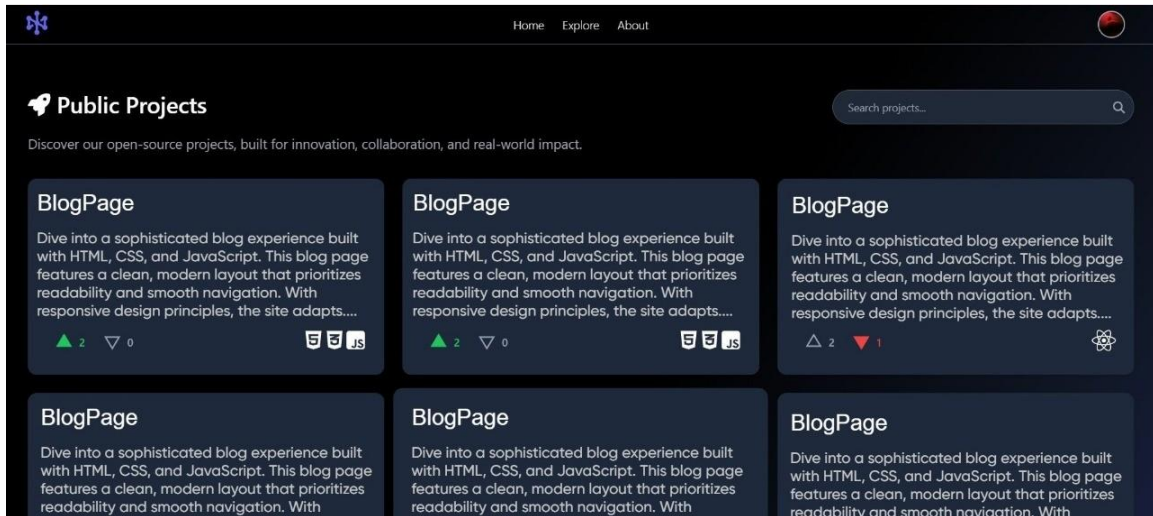
## ◆ Landing Page



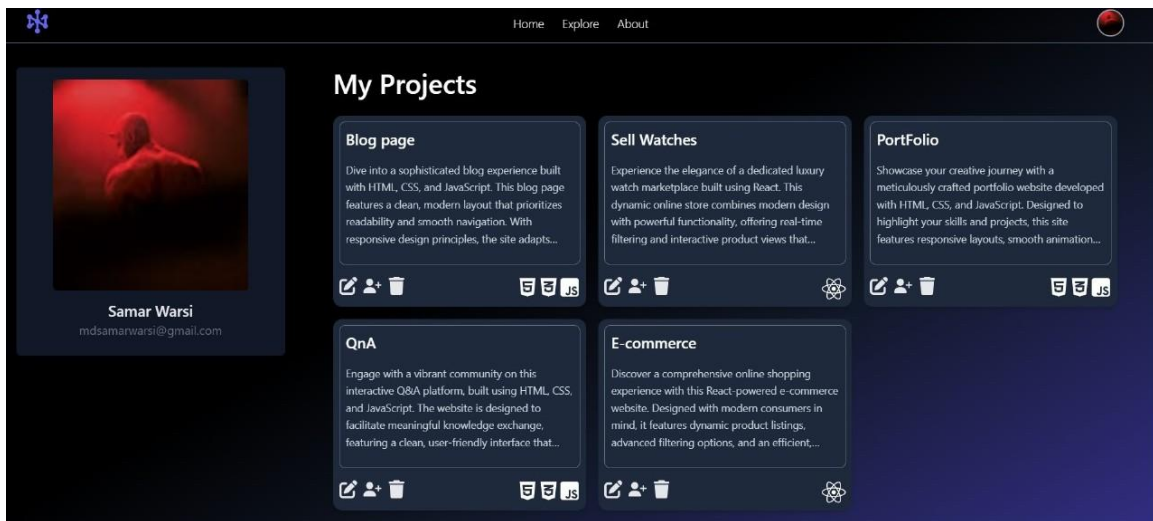
## ◆ Prompt Space



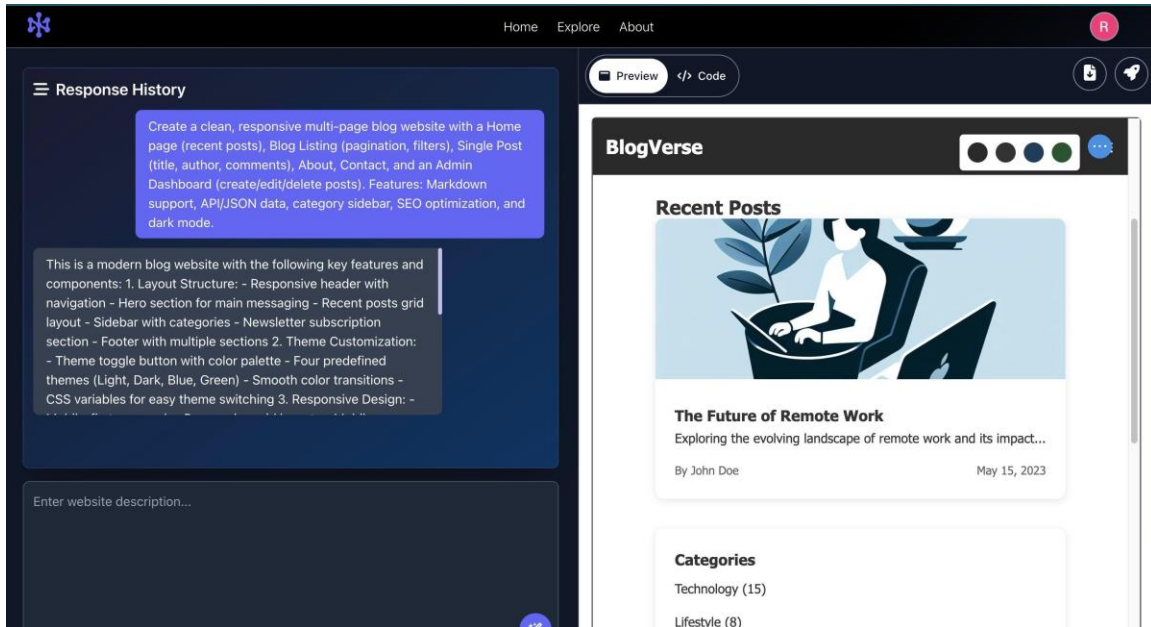
## ◆ Explore Page



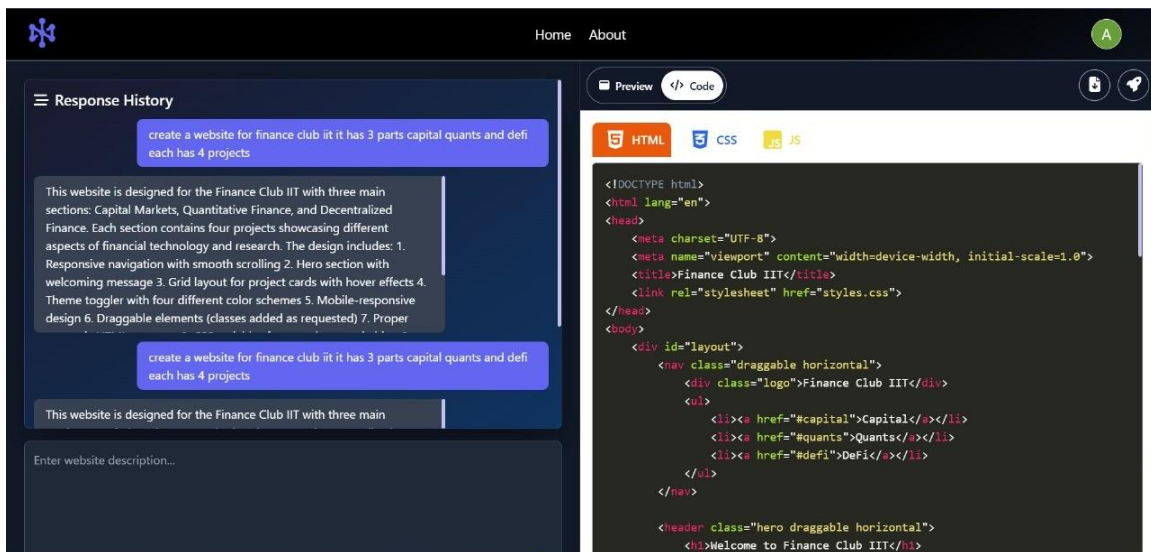
## ◆ Profile Page



## ◆ Generated Portfolio Preview



## ◆ Generated Code View



## 8. Challenges Faced

- Integration with Claude API required prompt engineering.
- Real-time collaborative editing synchronization issues.
- Drag-and-drop UX implementation complexities.

## 9. Learnings and Impact

- Practical experience with full-stack technologies.
- Hands-on with OAuth 2.0 and session handling.
- Full understanding of frontend-backend synergy.

## 10. Future Scope

- Resume parsing
- Version history
- Industry-specific templates
- Portfolio analytics

## 11. Team Members

- Vibha – 29 (<https://github.com/Vibha17>)
- Aryan - 03 (<https://github.com/aryanthepain>)
- Tejas - 27 (<https://github.com/tejas>)
- Vaishnavi – 28 (<https://github.com/VA0910>)

## 12. Conclusion

HexCode simplifies the way portfolios are created—bridging creativity and automation. By using AI to convert human input into aesthetically pleasing websites, it offers a powerful tool for anyone looking to build an online presence with minimal effort.